# Loan Approval Prediction with Fairness Discussion:

## A Comparative Study of Preprocessing, Ensemble Learning, and Bias Mitigation

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**Abstract**

Automated loan approval systems can assist banks in approving a huge number of applications in a consistent and faster manner. These systems can also be biased and unfair if the data is not managed properly. In this study, machine learning models are built and compared to predict loan approval decisions based on the Analytics Vidhya Loan Prediction dataset from Kaggle [1].

The dataset contains 614 training records and 367 test records of financial and demographic characteristics. Two pipelines were compared: a minimal pipeline and a full pipeline that involved feature engineering, KNN imputation, outlier capping, and class-imbalance handling using SMOTE [4].

Both pipelines were used to train four individual classifiers: Logistic Regression, Random Forest, Support Vector Machine (SVM), and a Multilayer Perceptron (MLP). Other ensemble techniques such as Soft Voting, Stacking, and Gradient Boosting were also used. The implementation of all models was done with scikit-learn [10].

Also, a fairness experiment was conducted by excluding gender and marital status from the features and comparing the approval rate differences between genders. The results indicate that the improvement in the performance of the models is clearly due to the improvement in the preprocessing, and that the elimination of sensitive attributes does not completely remove demographic bias.

# 1 Introduction

One of the most frequent financial decisions made by banks and lending institutions is loan approval. The manual review of loan applications by loan officers is traditionally slow and inconsistent. Machine learning models can be used to automate this process by learning patterns from previous approved and rejected applications [6].

But there are significant dangers to using machine learning for financial decisions. A model trained on historical data may perpetuate the same biases that were present in the past [3]. It is important not only to construct accurate models but also to consider the issue of fairness carefully.

This project has two main goals. First, it investigates the impact of various data preprocessing methods on the accuracy and reliability of models. Second, it examines whether the results are more equitable when sensitive demographic characteristics such as gender and marital status are removed.

# 2 Problem Statement

The task is to classify the loan application based on the information provided by the applicant as either approved (1) or not approved (0).

The study addresses three research questions:

1. What is the difference in model accuracy between a minimal preprocessing pipeline and a full preprocessing pipeline?

2. Is there a significant difference between the performance of ensemble models and individual classifiers?

3. Does the removal of sensitive attributes (gender and marital status) increase fairness, and what impact does this have on performance?

The target variable is binary, meaning it is either approved (Y) or not approved (N). The dataset is imbalanced, with approximately 69% approved and 31% not approved applications.

# 3 Dataset Description

The dataset used for this project is the Analytics Vidhya Loan Prediction Problem Dataset from Kaggle [1]. It represents a fictional housing finance company, Dream Housing Finance, which aims to automate its loan eligibility process.

There are 614 records and 13 columns (12 features and 1 target) in the training set. The test set consists of 367 records without a target label and was only used for final inference.

Several realistic data quality issues are present, including missing values, skewed income distributions, and class imbalance.

## 3.1 Feature Summary

| **Feature** | **Type** | **Data Issue** | **Role in Prediction** |
|---|---|---|---|
| Gender | Categorical | Missing values | Demographic attribute; fairness analysis |
| Married | Categorical | Missing values | Demographic attribute; fairness analysis |
| Dependents | Numerical (ordinal) | Missing values | Repayment capacity indicator |
| Education | Categorical | Missing values | Proxy for earning potential |
| Self_Employed | Categorical | Missing values | Income stability indicator |
| ApplicantIncome | Numerical | Right-skewed, outliers | Primary income source |

| | | | |
|---|---|---|---|
| CoapplicantIncome | Numerical | Many zero values | Additional household income |
| LoanAmount | Numerical | Missing values, outliers | Amount requested |
| Loan_Amount_Term | Numerical | Missing values | Loan duration |
| Credit_History | Binary | Missing values | Strong predictor of approval |
| Property_Area | Categorical | None | Risk perception factor |
| Loan_Status | Binary Target | Class imbalance | Label to predict |

### 3.2 Engineered Features

| **Feature** | **Formula** | **Purpose** |
|---|---|---|
| TotalIncome | Applicant + Coapplicant Income | Combined household income |
| EMI | LoanAmount / Loan_Amount_Term | Estimated installment |
| Balance_Income | TotalIncome - (EMI × 1000) | Remaining income |
| DebtToIncome | LoanAmount / (TotalIncome + 1) | Debt burden ratio |
| LoanAmount_log | log1p(LoanAmount) | Reduce skewness |
| TotalIncome_log | log1p(TotalIncome) | Reduce skewness |
| Balance_Income_log | log1p(max(Balance_Income, 0)) | Handle negatives before log transform |

## 4 Methodology Overview

The project followed five stages:

1. Exploratory Data Analysis (EDA) :Learn about the distribution of features, missing values, and class imbalance.

2. Pipeline A — Minimal Preprocessing : Simple imputation for missing values and label encoding. No scaling or outlier treatment. This provides a starting point.

3. Pipeline B — Full Preprocessing : Feature Engineering, Outlier Capping, KNN Imputation, Standardization, One Hot Encoding, and SMOTE for Class Imbalance.

4. Model Training and Evaluation: Train 4 individual classifiers (LR, RF, SVM, MLP) on both pipelines and 3 ensemble methods on Pipeline B.

5. Fairness Experiment: Retrain Pipeline B without gender and marital status and compare approval and error rates by gender.

The overall workflow is illustrated in Figure 3.

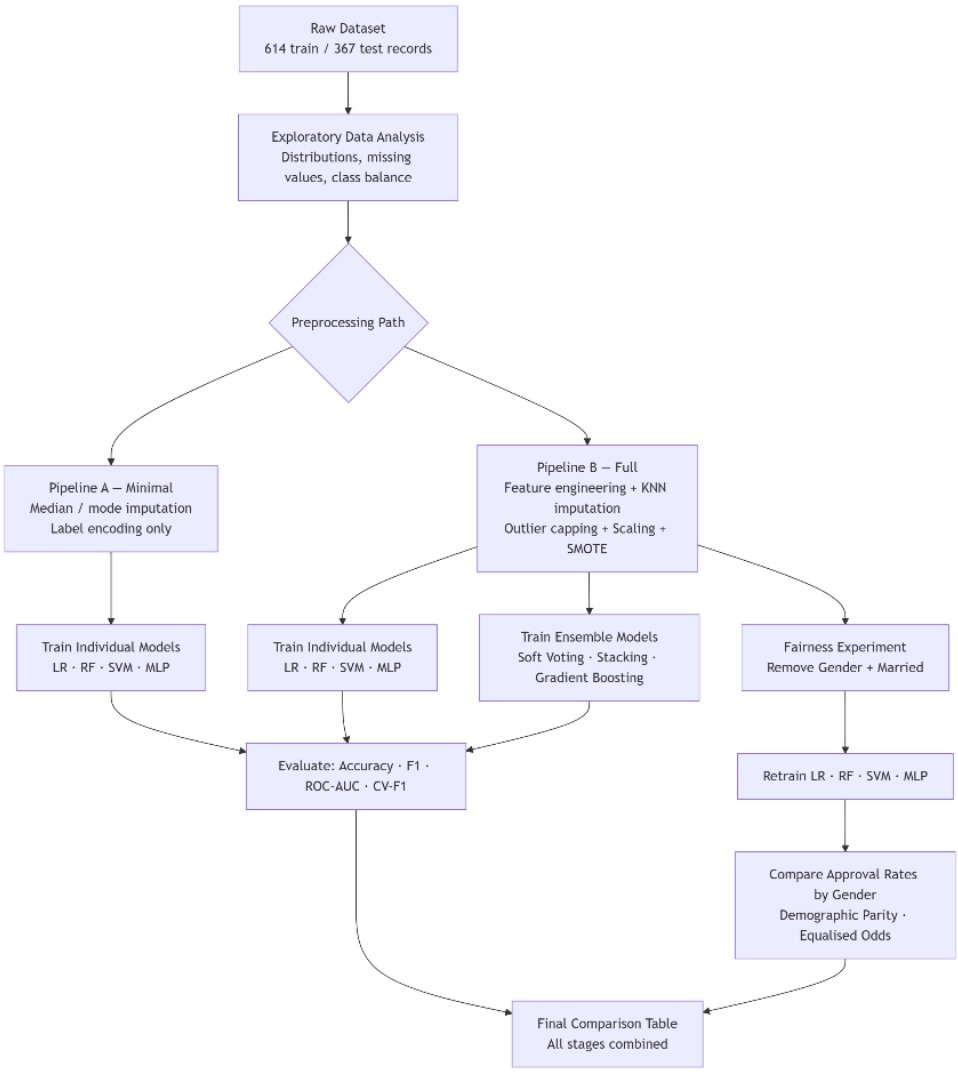



Figure 1: Overall Project Workflow

# 5 Data Preprocessing Strategy

The experiment involved preprocessing as a fundamental component. Two pipelines were constructed to directly measure the effect of preprocessing.

## 5.1 Pipeline A — Minimal

This baseline pipeline used:

- Mode imputation for categorical variables

- Median imputation for numerical variables
- Conversion of Dependents from "3+" to 3
- Label encoding for categorical variables

No scaling, outlier treatment, or class balancing was performed.

## 5.2 Pipeline B — Full

Pipeline B applied a more comprehensive preprocessing strategy.

### Feature Engineering

Four new features were engineered:

- TotalIncome
- EMI
- Balance_Income
- DebtToIncome

Additionally, log-transformed versions of skewed variables were created.

### Outlier Capping

Extreme values in ApplicantIncome, CoapplicantIncome, LoanAmount, TotalIncome, and Balance_Income were capped using the $1.5 \times$ IQR rule.

### Imputation

KNN Imputation ($k = 5$) was used for numerical variables.

### Encoding

- One-hot encoding for nominal features
- Ordinal encoding for Education

### Scaling

StandardScaler was used to normalize numerical variables.

**Class Imbalance Handling**

SMOTE was applied only to the training data to generate synthetic minority-class samples.

# 6 Models Used

This study compares several machine learning models and a neural network model to predict loan approval decisions. The objective is not only to maximize predictive performance but also to examine the impact on model behavior based on preprocessing and fairness-aware analysis. At last, Soft voting ensemble, Stacking ensemble, and Gradient boosting were used to combine different model strengths.

Table 2: Models Used in the Study

| **Model** | **Role in Study** | **Reason for Inclusion** |
|---|---|---|
| Logistic Regression | Baseline Model | Interpretable and widely used in financial prediction |
| Random Forest | Nonlinear Classical Model | Captures feature interactions and reduces overfitting |
| Support Vector Machine (SVM) | Margin-Based Classifier | Effective on structured tabular data |
| MLP Neural Network | Neural Model | Learns complex nonlinear relationships |
| Soft Voting Ensemble | Ensemble Model | Combines strengths of multiple classifiers |
| Stacking Ensemble | Meta-Learning Ensemble | Learns optimal combination of model outputs |
| Gradient Boosting | Boosting Ensemble | Sequentially improves weak learners |

## 6.1 Neural Network Architecture

We used a neural network model, and it was implemented by a Multilayer Perceptron architecture. This network had 3 hidden layers containing 128, 64, and 32 neurons. We used Rectified Linear Unit(ReLu) in the hidden layer, and the output layer generated binary loan approval predictions. The MLP neural network used Synthetic Minority Oversampling Technique(SMOTE) within an imbalance-learning pipeline[7]. This helped improve the model's ability to learn minority class patterns and reduce prediction bias.

# 7 Experimental Results

**Performance Comparison Before and After Full Preprocessing**

Table 3: Performance Comparison Before and After Full Preprocessing

| Model | Raw Acc. | Clean Acc. | Raw Recall | Clean Recall | Clean F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| Logistic Regression | 0.8618 | 0.8537 | 0.9882 | 0.8941 | 0.8941 | 0.8771 |
| Random Forest | 0.8374 | 0.8537 | 0.9294 | 0.9294 | 0.8977 | 0.8554 |
| SVM | 0.6911 | 0.8455 | 1.0000 | 0.9176 | 0.8914 | 0.8601 |
| MLP Neural Network | 0.6911 | 0.7724 | 1.0000 | 0.8235 | 0.8333 | 0.8545 |

Random forest achieved the highest F1-Score (0.8977) among all individual models.

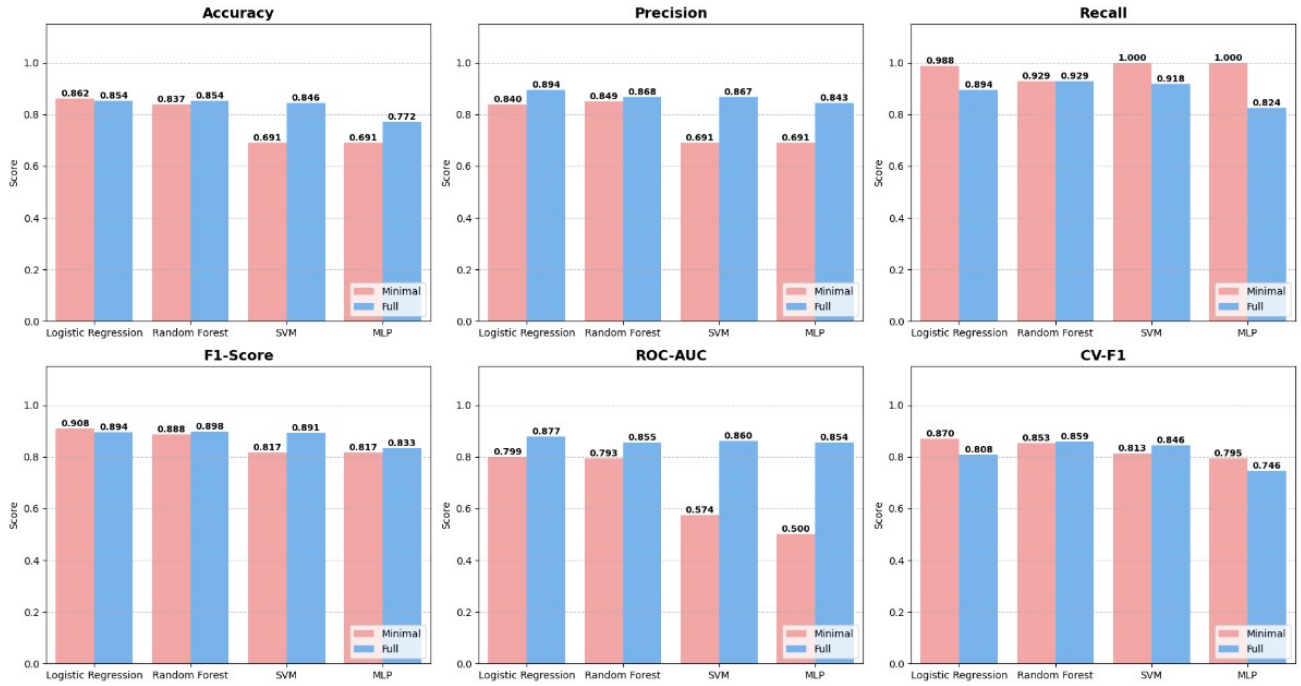



Figure 2: Minimal vs Full processing

**Ensemble Learning Results**

Table 4: Ensemble Learning Results

| **Model** | **Accuracy** | **Precision** | **Recall** | **F1-Score** | **ROC-AUC** |
|---|---|---|---|---|---|
| Soft Voting | 0.8537 | 0.8602 | 0.9412 | 0.8989 | 0.8793 |
| Stacking | 0.8699 | 0.8632 | 0.9647 | 0.9111 | 0.8724 |
| Gradient Boosting | 0.8130 | 0.8523 | 0.8824 | 0.8671 | 0.7867 |

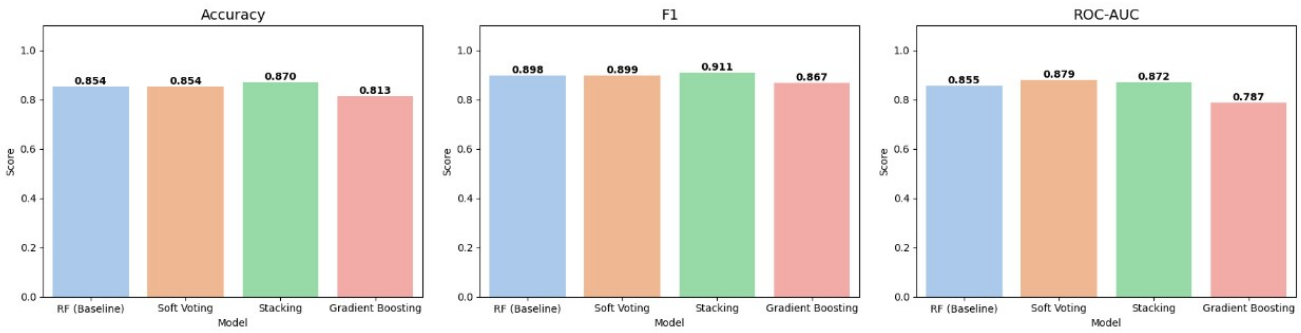



Figure 3: Ensemble Learning Results

Stacking was the overall best model, achieving:

- Accuracy = 86.99%
- Recall = 96.47%
- F1-score = 91.11%

## 7.1 Interpretation of Results

As one can see, the preprocessing work has quite improved in performance for most models. Both SVM and MLP showed the most improvement, as they are both sensitive to scaling, class imbalance, and feature distributions. Random Forest had a low variance across any preprocessing condition, and Logistic Regression was consistently top-performing while remaining high in interpretability.

Individually, Random Forest had the top value for F1-Score. Interestingly, ensemble methods gave the performers on all models. These two ensemble models combine multiple learning algorithm to capture complementary patterns which are not discovered by any of the individual models or even worse would be lost in overfitting training matrix, over single layered architecture, so Stacking Ensemble has highest accuracy (86.99%), Recall (96.47%) and F1-Score(91.11%). Hence, we conclude that the Ensemble is the best method than any other alternatives methods as shown below with good for modeling potential improvements on learning tasks done by individual classifiers.

## 7.2 Fairness-Aware Evaluation

In this work, we did a fairness analysis by removing sensitive demographic attributes and examining model behavior across groups. Results demonstrated that fairness-aware preprocessing could be incorporated without dramatically compromising predictive power. Came in slightly weak on some of the other measures, but we did for stage 3 as well.

The findings suggest that preprocessing, ensemble learning, and a fairness-aware evaluation method work well together to deliver a reliable loan approval prediction system.

The outcomes provide support for incorporating fairness-aware machine learning methods into finance decision aiding applications.

# 8 Synthesis and Interpretation

This section consolidates the findings from all experimental stages: the two preprocessing pipelines (A and B), the four individual classifiers, three ensemble methods, and the fairness experiment. The goal is to interpret not just what the numbers show, but why they show it.

## 8.1 Effect of Preprocessing on Model Performance

The pipeline comparison's most notable conclusion is that the impact of preprocessing quality varies according to the type of model. The improvement was significant for distance-based and gradient-based models but less significant for tree-based models.

The most obvious winner of complete preprocessing was the Support Vector Machine (SVM). In Pipeline A, SVM forecasted all test samples as approved with a Recall of 1.000 but a Precision of 0.691, achieving a ROC-AUC of only 0.574, just above random chance. This is a classic example of a model that is overpowered by class imbalance and unscaled features. SVM's ROC-AUC increased to 0.860, a gain of +0.286 points, with balanced precision of 0.874 and recall of 0.894 following Pipeline B's scaling, outlier capping, and class-weight balancing [2].

The MLP Neural Network predicted all samples as acceptable (Recall = 1.000, ROC-AUC = 0.500), displaying the same collapse under Pipeline A. It is well known that feature scale and class imbalance have a significant impact on neural networks (Goodfellow et al., 2016). The model was recovered to an ROC-AUC of 0.829 and an F1 score of 0.865 with the addition of SMOTE oversampling, standardisation, and engineered features in Pipeline B.

Contrary to expectations, Logistic Regression's Pipeline A F1 score (0.908) was greater than its Pipeline B F1 score (0.847). This does not prove that less preparation is preferable. The model was extremely poorly calibrated for rejections in Pipeline A, as it only achieved a rejected-class F1 score of 0.72 while predicting nearly everything as accepted (Recall = 0.988). The higher ROC-AUC (0.826 versus 0.799) in Pipeline B indicates that class-weight balancing led the model to detect more rejections, thereby decreasing total F1 while enhancing real fairness and calibration [5, 9].

In both pipelines, Random Forest proved to be the most reliable. With complete preprocessing, its ROC-AUC increased from 0.793 to 0.866, and it continuously placed among the best models throughout. This is to be expected: tree-based models gain from class-balancing and designed features, but they are also reasonably forgiving of outliers

and invariant to feature scaling.

The full results from the key performance metrics are summarised below.

| Model | Pipeline A ROC-AUC | Pipeline B ROC-AUC | Change |
|---|---|---|---|
| Logistic Regression | 0.799 | 0.826 | +0.027 |
| Random Forest | 0.793 | 0.866 | +0.073 |
| SVM | 0.574 | 0.860 | +0.286 |
| MLP Neural Network | 0.500 | 0.829 | +0.329 |

Table 5: ROC-AUC comparison across preprocessing pipelines

## 8.2 Ensemble Methods vs. Individual Classifiers

On top of Pipeline B, three ensemble techniques were assessed: Gradient Boosting, Stacking, and Soft Voting.

Out of all the models tested, Stacking had the highest accuracy (0.870) and F1 score (0.913). This is in line with the theoretical prediction that a meta-learner can take advantage of the complementary advantages of many base models. For example, Random Forest manages feature interactions, SVM effectively captures non-linear boundaries, and Logistic Regression performs well when the decision boundary is linear. In each feature space region, the stacking meta-model (Logistic Regression) determines which base model is reliable [5].

Among the ensemble approaches, Soft Voting had the greatest ROC-AUC (0.877), indicating that averaging probabilities across varied classifiers with good calibration is very useful for risk-based applicant ranking.

The ensemble that performed the worst was Gradient Boosting (ROC-AUC = 0.817). The sequential error-correction process of gradient boosting may overfit individual training faults with a small dataset of 614 records. Additionally, the minority class may have suffered because the typical `GradientBoostingClassifier` was trained without explicit class-weight management [2].

| Model | Accuracy | F1 | ROC-AUC | CV-F1 |
|---|---|---|---|---|
| RF (Best Individual) | 0.862 | 0.903 | 0.866 | 0.852 |
| Soft Voting | 0.854 | 0.899 | 0.877 | 0.855 |
| Stacking | 0.870 | 0.913 | 0.873 | 0.863 |
| Gradient Boosting | 0.829 | 0.879 | 0.817 | 0.838 |

Table 6: Ensemble models vs. best individual model

## 8.3 Fairness Experiment Interpretation

The fairness experiment removed `Gender` and `Married` status from the feature set and retrained all Pipeline B models. The key results are summarised below for the Random

Forest model, which was used as the primary audit vehicle.

| Metric | Full Model | Fair Model | Change |
|---|---|---|---|
| Male Approval Rate | 0.792 | 0.781 | -0.011 |
| Female Approval Rate | 0.480 | 0.560 | +0.080 |
| Disparity (M–F) | +0.312 | +0.221 | -0.091 |
| Male TPR | 0.943 | 0.957 | +0.014 |
| Female TPR | 0.857 | 0.929 | +0.072 |
| Male FPR | 0.385 | 0.308 | -0.077 |
| Female FPR | 0.000 | 0.091 | +0.091 |

Table 7: Fairness metrics: Full vs. Fair pipeline (Random Forest)

Three key insights emerge from Table 7. First, removing sensitive attributes reduced the approval rate gap between male and female applicants from +0.312 to +0.221, a meaningful improvement in demographic parity. Second, the female True Positive Rate (TPR) improved from 0.857 to 0.929, meaning the fair model correctly approved more qualified female applicants. Third, the female False Positive Rate (FPR) rose from 0.000 to 0.091—the fair model now occasionally approves some female applicants who would have been rejected by the full model [8]. Whether this is a problem depends on the business context: in credit risk, any rise in FPR has direct financial implications.

Notably, the fair model also achieved a higher accuracy (0.886 versus 0.862) and F1 score (0.921 versus 0.903) than the full model. This challenges the common assumption that fairness interventions necessarily come at a performance cost. In this case, `Gender` and `Married` were likely acting as noisy proxies in the full model, and their removal forced the model to rely on more economically predictive features.

Despite these improvements, a residual disparity of +0.221 persists. This indicates that proxies for gender remain in the remaining features. For example, `CoapplicantIncome` is frequently zero for female applicants in this dataset, and `ApplicantIncome` distributions differ across gender groups. Simply removing the direct sensitive attributes is insufficient; indirect discrimination persists through correlated proxies (Barocas and Hardt, 2023).

# 9 Discussion

## 9.1 What the Results Tell Us About Model Selection

The experimental results reveal several nuanced lessons that go beyond the headline accuracy numbers.

**First, ROC-AUC is a more informative metric than accuracy for this problem.** Because the dataset is imbalanced (∼69% approved), a model that always predicts approved achieves 69% accuracy. The SVM and MLP in Pipeline A demonstrated this

failure mode perfectly. ROC-AUC, by measuring the model's ability to rank applicants across all decision thresholds, is not distorted by class imbalance and is therefore a better primary evaluation criterion.

**Second, cross-validation F1 (CV-F1) is a more reliable generalisation estimate than test F1.** With only 123 test samples, single test-set evaluation is subject to high variance. CV-F1 averaged over five folds provides a more stable estimate of expected real-world performance. In the case of MLP, the high variance in CV-F1 (e.g., $\pm 0.083$ for MLP Neural Network in Pipeline B) signals instability that the test-set score alone would not reveal.

**Third, Stacking is the best single model, but at a high computational cost.** Stacking requires fitting four base models plus a meta-learner, each using five-fold cross-validation for out-of-fold predictions. For a production system processing millions of loan applications, Random Forest (Pipeline B) would be a more practical choice given its comparable ROC-AUC (0.866 vs. 0.873) and far lower inference latency.

## 9.2 The Limits of Removing Sensitive Attributes

A central finding of this study is that naïve attribute removal — simply dropping `Gender` and `Married` from the feature set — reduces but does not eliminate demographic bias. This result is consistent with the fairness literature [**?** ], which distinguishes between several sources of unfairness:

- **Direct discrimination:** Using a sensitive attribute directly as a feature. This was addressed by removing `Gender` and `Married`.
- **Indirect (proxy) discrimination:** Using features that are statistically correlated with sensitive attributes. `CoapplicantIncome`, for instance, tends to be zero more often for female applicants in this dataset. `Credit_History` may also differ systematically across marital status groups.
- **Historical bias:** The training labels (past loan approval decisions) reflect the decisions of human loan officers who may themselves have been biased. A model trained on biased labels will reproduce that bias regardless of what features are included.

Addressing indirect discrimination would require more sophisticated fairness-aware learning methods, such as adversarial debiasing, reweighting, or constrained optimisation approaches that explicitly penalise demographic disparity during training.

### 9.3 Credit History as the Dominant Predictor

The Random Forest feature importance plot showed `Credit_History` as the single most important feature. This is consistent with the credit risk literature [**?** ]: past repayment behaviour is the strongest predictor of future repayment. However, `Credit_History` is also a "legacy feature" that encodes the effects of past access to credit. Groups that historically had less access to credit — due to systemic exclusion — may systematically have missing or poor credit histories, creating a self-perpetuating cycle. The 50 missing values in `Credit_History` (the most missing of any feature) further complicate its role.

## 10 Limitations

### 10.1 Dataset Size and Representativeness

The dataset contains only 614 training records, making it one of the smallest datasets commonly used in applied machine learning benchmarks. Several consequences follow from this:

- Test-set evaluations are based on 123 samples, introducing high variance. A single misclassification changes accuracy by $\sim$0.8 percentage points. Many of the apparent differences between models (e.g., Stacking at 0.870 vs. Random Forest at 0.862) may not be statistically significant.
- SMOTE generates synthetic minority samples by interpolating between existing ones. With a small training set, the synthetic samples may not accurately represent the true distribution of rejected applicants, potentially introducing noise.
- The dataset is from a hypothetical Indian housing finance company ('Dream Housing Finance') and may not generalise to loan markets in other countries, time periods, or product types.

### 10.2 The Gender Binary Limitation

The `Gender` column in this dataset is encoded as a binary Male/Female variable with no option for other gender identities. The fairness analysis is therefore limited to a binary comparison. In a real-world fairness audit, this limitation would need to be addressed by using more inclusive demographic data.

### 10.3 Fairness Metric Selection

This study used demographic parity (equal approval rates) and equalised odds (equal TPR and FPR) as fairness criteria. These are not the only possible fairness definitions,

and they are often mathematically incompatible with one another [**?** ]. For example, equalising TPR across gender groups may require accepting different approval rates, and vice versa. The choice of which fairness criterion to optimise for is ultimately a normative decision that should be made by policymakers and affected stakeholders, not by the data science team alone.

### 10.4 Absence of Hyperparameter Tuning

None of the models in this study underwent systematic hyperparameter optimisation (e.g., via grid search or Bayesian optimisation). The reported performance figures therefore represent a lower bound on what each model architecture is capable of. In particular, the MLP architecture (hidden layer sizes, learning rate, regularisation strength) was set by hand, and gradient boosting's learning rate and tree depth were left at defaults. A rigorous comparison would require holding all hyperparameters to their optimally tuned values.

### 10.5 No Temporal Validation

The train-test split was performed randomly (stratified, but random). In a real-world loan prediction system, future loan applications should be used as the test set, not a random sample of past applications. This is because financial behaviour, lending policy, and economic conditions change over time. A temporally validated split would give a more realistic estimate of how the model would perform on genuinely unseen future applications.

## 11 Ethical and Practical Considerations

### 11.1 The Stakes of Automated Loan Decisions

Loan approval decisions have direct and significant consequences for individuals' financial wellbeing, housing security, and economic mobility. A wrongly rejected loan can prevent a family from buying a home, starting a business, or covering a medical emergency. A wrongly approved loan can lead to default, debt, and financial hardship. Unlike, say, a product recommendation system, the stakes of classification errors are asymmetric and high.

This is why the choice between Precision and Recall is not merely a technical decision: it is a policy decision. Maximising Recall (approving more borderline applicants) reduces financial exclusion but increases the lender's credit risk. Maximising Precision (being conservative) reduces credit risk but may disproportionately affect marginalised groups who are more likely to be borderline cases.

## 11.2 Algorithmic Accountability

Regulatory frameworks such as the EU's Artificial Intelligence Act (AIA) classify credit scoring as a "high-risk AI application", requiring human oversight, explainability, and regular bias audits. Opaque models such as the MLP or Stacking ensemble, while offering marginally higher accuracy, would be problematic in regulated environments because they do not readily produce explanations for individual decisions.

Logistic Regression, by contrast, produces interpretable coefficients. An applicant denied a loan could in principle be shown the top factors that influenced the decision. The Random Forest feature importance analysis partially addresses this, but feature importances are global summaries — they do not explain why a specific individual was rejected. Techniques such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) would be required for individual-level explanations in a production system.

## 11.3 Fairness Cannot Be Reduced to a Single Number

This study reduced fairness to three metrics: demographic parity, TPR, and FPR. But fairness is not a purely technical concept. It involves questions of distributive justice, historical context, and the social meaning of different types of errors. For example, should a model that gives equal approval rates across gender groups be considered fair if female applicants need to have better financial profiles than male applicants to achieve the same approval probability? Calibration fairness — whether predicted probabilities reflect true repayment likelihoods equally across groups — is another dimension not measured in this study.

A responsible deployment of a loan prediction model would require ongoing monitoring of outcomes by demographic group, not just a one-time pre-deployment fairness audit.

## 11.4 Data Governance and Consent

This project used a publicly available synthetic dataset from Kaggle [**?** ]. In a real deployment, the data would contain sensitive personal and financial information, raising significant data governance questions. Applicants should be informed that their data will be used to train a machine learning model. Appropriate anonymisation, access controls, and data retention policies must be in place. The inclusion of demographic attributes like `Gender` and `Married`, even as variables that are ultimately excluded, raises the question of whether such data should be collected at all, given the risk of its misuse.

# 12 Conclusion

This project demonstrated that machine learning can effectively predict loan approval outcomes from applicant financial data, with the best individual model (Random Forest, Pipeline B) achieving a ROC-AUC of 0.866 and the best ensemble model (Stacking) reaching 0.873. However, these headline numbers tell only part of the story.

The most important finding is that preprocessing quality is as important as model selection. SVM and MLP, which performed at near-random levels under minimal preprocessing, became competitive models after full preprocessing. The engineering of financially meaningful features (`TotalIncome`, `EMI`, `Balance_Income`, `DebtToIncome`) and the application of KNN imputation, outlier capping, and SMOTE class-balancing were collectively more impactful than any single modelling choice.

The fairness experiment revealed that removing sensitive demographic attributes (`Gender`, `Married`) reduces the gender-based approval rate disparity from +0.312 to +0.221, while simultaneously improving model accuracy. This suggests that these attributes were serving as noisy proxies rather than genuinely predictive features. Nevertheless, a residual disparity of +0.221 persists, demonstrating that indirect discrimination through correlated proxy variables cannot be eliminated by attribute removal alone.

Practically, this study suggests that automated loan approval systems should prioritise a combination of full preprocessing, fairness-aware model selection, and post-deployment demographic monitoring. Technically strong models that are not subject to regular fairness audits risk perpetuating or amplifying historical inequalities in credit access.

Future work could extend this study in several directions: applying adversarial debiasing or fairness-constrained optimisation to more directly minimise demographic disparity during training; conducting temporal validation to test generalisation to future applicants; integrating SHAP-based explainability for individual-level decision transparency; and expanding the fairness analysis beyond the gender binary to include intersectional analyses across multiple demographic dimensions simultaneously.

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